

BITS PILANI, DUBAI CAMPUS
Dubai International Academic City, Dubai.
Third Year - II Semester 2018-19
Design and Analysis of Algorithms (CS F364)
Test -1 (Closed Book)

Date: 26.02.2019

Max Marks: 40

Time: 2.30 to 3.20 am, 50 minutes

Weightage: 20%

1. Justify the choice of pivot which gives the lowest running time complexity for Quicksort? Also Which pivot choice gives the highest time complexity of Quicksort and why? Justify both pivot choices through best case, worst case and average case computational complexity analysis.

[7]

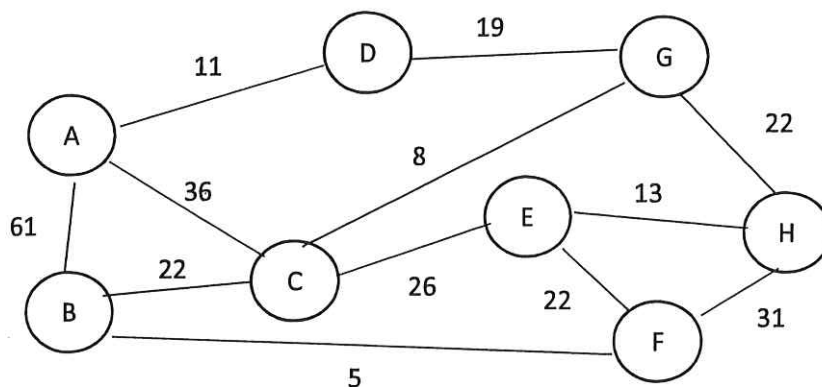
2. Can we improve the Time Complexity of Insertion Sort by using Binary Search? Imagine a variation of Insertion Sort, called BinaryInsertionSort which uses Binary Search to insert new elements in the correct position in the sorted portion of the array. Write Pseudocode for BinaryInsertionSort and justify whether it improves the asymptotic time complexity of Insertion Sort or not (Best Case, Worst Case, Average Case)?

[7]

3. Write pseudocode for computing the kth-maximum element of an array. It is given that all elements of the array are distinct. What is the Best Case, Worst Case and Average Case time complexity of your algorithm? Does it depend on k? Why, or why not? Write the most efficient algorithm.

[6]

4. For the following given graph, it is required to construct Maximum Spanning Tree. The spanning tree constructed which maximizes the total cost all edges. Suggest the modification in Prim's Algorithm to construct Maximum Spanning Tree. Write the algorithm for modified Prim's Algo for maximum spanning tree. For the following given graph construct Maximum Spanning Tree using Prim's method.



[7]

5. Following jobs are given with deadline and energy required. Perform job sequencing without violating the deadline. But total energy consumed is to be minimized. An i^{th} job associated with deadline $d_i > 0$. Each job has to be completed before deadline. At least 4 jobs have to be completed. The number of jobs $n = 6$. Determine the job sequence using Greedy method.

Job	1	2	3	4	5	6
Deadline	4	1	2	3	1	2
Energy Required	13	8	10	9	18	5

[7]

6. Write the algorithm for Tree Vertex Split using Greedy Method.

[6]

Name:

ID Num:

Faculty Name:

BITS PILANI, DUBAI CAMPUS
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Third Year - II Semester 2018-19
Design and Analysis of Algorithms (CS F364)
Quiz -1 (Closed Book)

Date: 11.03.2019

Max Marks: 20

Time: 11.10 to 11.30 am, 20 minutes

Weightage: 10%

1. Answer the following statement is True /False

a. Dynamic programming is an algorithm design method that can be used when a solution to the problem is viewed as the result of sequence of decisions.

Ans:

b. Dynamic Programming definitely gives the optimal solution always.

Ans:

c. In Dynamic Programming, Decision sequences containing subsequences that are sub optimal are not considered.

Ans:

d. For knapsack 0/1 problem the solution can be obtained by randomly picking the 0 or 1s.

Ans:

[4]

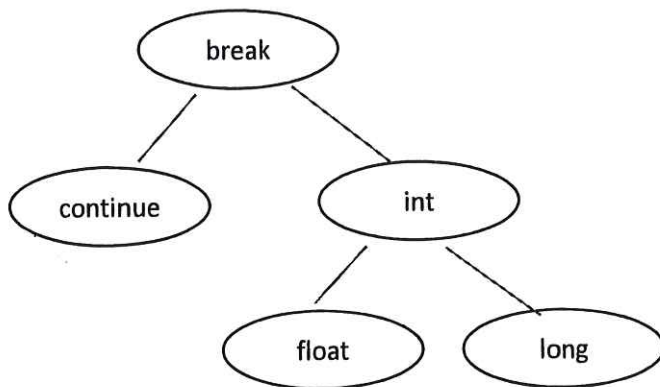
2. Define the Principle of Optimality

[3]

3. It is required to formulate All Pair shortest Path. Write the equation for shortest path from i to j going through no vertex higher than k.

[3]

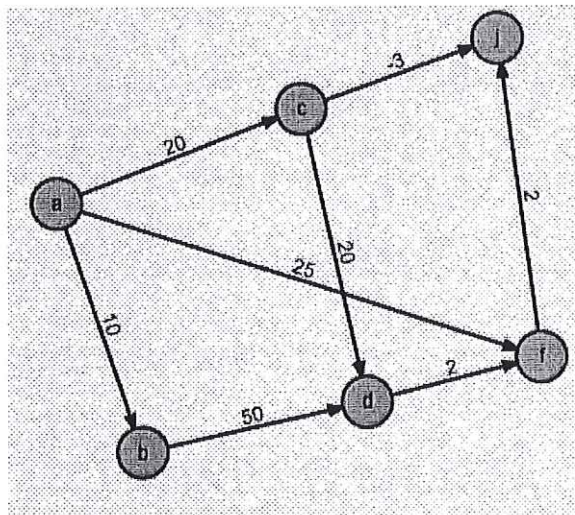
4. Consider five keywords (a1,a2,a3,a4,a5)=(int, float, continue, break, long) Assume equal probability for both internal and external nodes. Compute the cost following search tree.



Ans:

[4]

5. It is required to compute cost matrix of all pair shortest path for the following given graph.



[3+3]

a. write the cost matrix all pair shortest path A^0

b. write the cost matrix all pair shortest path A^2

BITS PILANI, DUBAI CAMPUS
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Third Year - II Semester 2018-19
Design and Analysis of Algorithms (CS F364)
Test -2 (Open Book)

Date: 10.04.2019

Max Marks: 40

Time: 2.30 to 3.20 am, 50 minutes

Weightage: 20%

1. It is required to construct Optimal Binary Search Tree. We have given 3 terms (long, continue, break). Here probability of successful search is $p(1:3)=(0.2, 0.14, 0.16)$ and probability of unsuccessful search $q(0:3)=(0.1, 0.15, 0.07, 0.18)$. Total search cost of binary tree is

$$\sum_{i=1}^n p_i \cdot \text{level}(a_i) + \sum_{i=0}^n q_i \cdot (\text{level}(\text{failure node } i) - 1)$$

Compute the optimal binary search cost $c(i, j)$ given as:

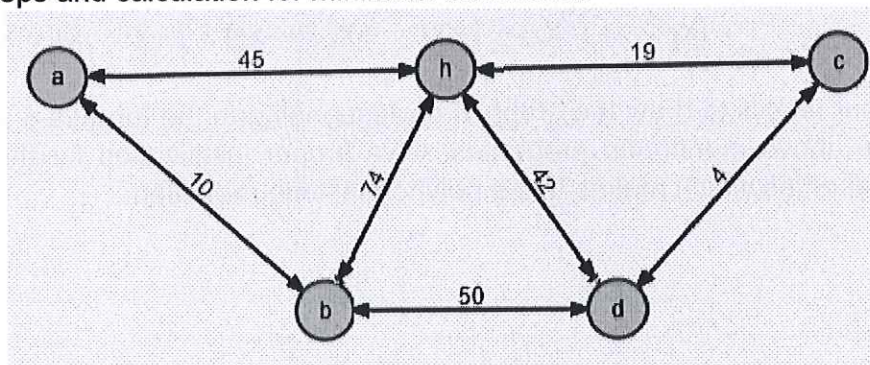
$$c(i, j) = \min_{i < k \leq j} \{c(i, k-1) + c(k, j)\} + w(i, j)$$

$$\text{Here } w(i, j) = p(j) + q(j) + w(i, j-1)$$

Write the calculated values in terms of a table rows ranging from 0 to 3 and columns 0 to 3. Show all the calculations.

[7]

2. For the given following graph, represent the graph in a matrix. Compute the minimum cost for travel sales person tour. The tour starts from the node 'a'. Show all the steps and calculation for minimum travel cost.



[7]

3. Write a non-deterministic algorithm to check whether a Boolean expression satisfies or not. Consider the Boolean expression consists of 'n' literals

[6]

4. Imagine a genie named Aladdin, who has the power to always guess correctly. Answer True or False with proper justification the following:

- a. In case P is a proper subset of NP , can the genie Aladdin compute something faster than ordinary humans who do not have his guessing power.
- b. In case P and NP are identical sets, can the genie Aladdin compute something faster than ordinary humans who do not have his guessing power.

[5]

5. Consider the following two problems:

- a. **Factor:** Given a number $n = pq$, the product of two distinct primes p and q , find p and q .
- b. **EulerTotient:** Given a number $n = pq$, where p and q are unknown primes, find $\phi(n) = (p - 1)(q - 1)$, the number of numbers less than n relatively prime to n .

Reduce Factor to EulerTotient. Is your reduction algorithm a polynomial time algorithm?

[8]

6. As part of polynomial time reductions, we transform the given 3-CNF formula ϕ into (G, k) such that $\phi \in 3SAT \Leftrightarrow (G, k) \in CLIQUE$. That is, the graph G should have a k -clique if and only if the ϕ is satisfiable; and the formula ϕ should be satisfiable if and only if the graph G has a k -clique. ϕ is given as the following 3-CNF formula:

$$(x_1 \vee x_1 \vee x_2) \wedge (\neg x_1 \vee \neg x_2 \vee \neg x_2) \wedge (\neg x_1 \vee x_2 \vee x_2)$$

Your task is to draw the graph G corresponding to the formula ϕ above that holds the above mentioned properties. Give proper justification for the graph and its properties which makes this a polynomial time reduction.

[7]

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Design and Analysis of Algorithms (CS F364)
Compre

Date: 28.05.2019

Time: 12.30 to 3.30 pm

Max Marks: 80

Weightage: 40%

PART A(40M)

1. For each algorithm listed below,

- a) Binary search
- b) Insertion sort
- c) Merge sort.

- give a recurrence that describes its worst-case running time, and
 - give its worst-case running time using Asymptotic notation
- Provide proper justification for your answers.

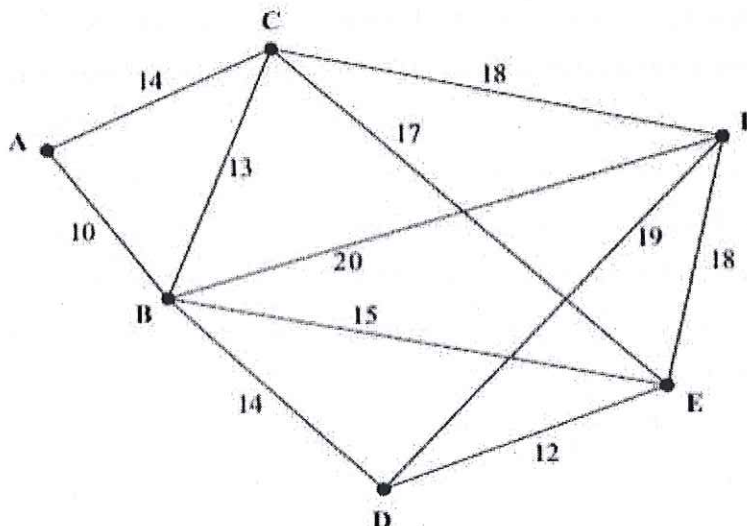
[7]

2. (a) Define the terms (i) tree, (ii) minimum spanning tree.

(b) Use Kruskal's algorithm to find a minimum spanning tree for the network shown in following figure. You should list the arcs (edges) in the order in which you consider them. In each case, state whether you are adding the arc to your minimum spanning tree.

(c) Draw your minimum spanning tree.

(d) State whether your minimum spanning tree is unique. Justify your answer.



[2+2+1+2=7]

3. State TRUE or FALSE, with proper justification:

- a) If A is an NP-complete problem, and problem B reduces (polynomially) to A, then B is an NP-complete problem.
- b) Imagine we have a computer with an infinite amount of RAM, in which any memory address can be accessed in constant time. There exists a computer program of infinite length that solves SAT in linear time. [Note: SAT is Boolean satisfiability problem determining if there exists an interpretation that satisfies a given Boolean formula]
- c) Poly-Time reducibility is an Equivalence Relationship. [Note: Equivalence relationship is reflexive, symmetric and transitive]
- d) Let F_k denote the k^{th} Fibonacci number. Then, the n^2 the Fibonacci number F_n^2 can be computed in $O(\lg n)$ time.
- e) Consider a modification of Quicksort, such that each time Partition is called, the median of the partitioned array is found (using a find median algorithm which works in $O(n)$ time) and used as a pivot. The worst-case running time of this algorithm is $\Theta(n \cdot \log n)$

[10]

4. Given a sequence of numbers $a_1, a_2, \dots, a_n$, we say that a_i is a local maximum if $a_i \geq a_{i+1}$, and $a_i \geq a_{i-1}$ (where the condition is ignored if $a_{i \pm 1}$ would be out of range).

- (a) Give an algorithm that given a sequence $a_1, a_2, \dots, a_n$ finds a local maximum and analyze its runtime. For full credit, your algorithm should run in time $O(\log(n))$.
- (b) What design techniques does your algorithm follow?

[8]

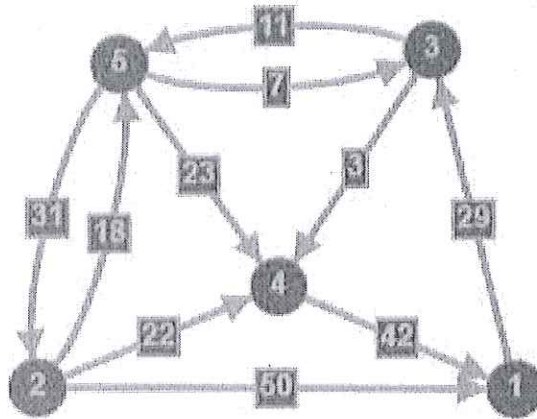
5. Consider the following computational problem. Given n points on the real line $x_1 < x_2 < \dots < x_n$, find a minimum number of unit length intervals $I_1, I_2, \dots, I_m$ that cover these points. In particular, each I_j should be an interval of the form $[y_j, y_j + 1]$ (unit length), and for every $1 \leq i \leq n$ the number x_i should be an element of I_j for some j (the intervals cover the points). Among such coverings, we are looking at one with m , the number of intervals, as small as possible.

- a) Is there a greedy algorithm that solves this problem? If yes, then provide the pseudocode for that greedy algorithm and prove it's correctness? If no, then justify why greedy does not work in this case.
- b) Let the points be 0, 0.3, 1.5, 2.1, 3. Find the optimal set of intervals that covers these points. Try applying your greedy algorithm here, if you came up with one.

[8]

PART B(40M)

1. Write the algorithm to find all pair shortest path in a given graph. For given following graph determine the cost matrix for all pair shortest. Stepwise execute the algorithm for all pair shortest path and compute the cost matrix for each iteration and write the cost matrix between A^0 to A^5 .



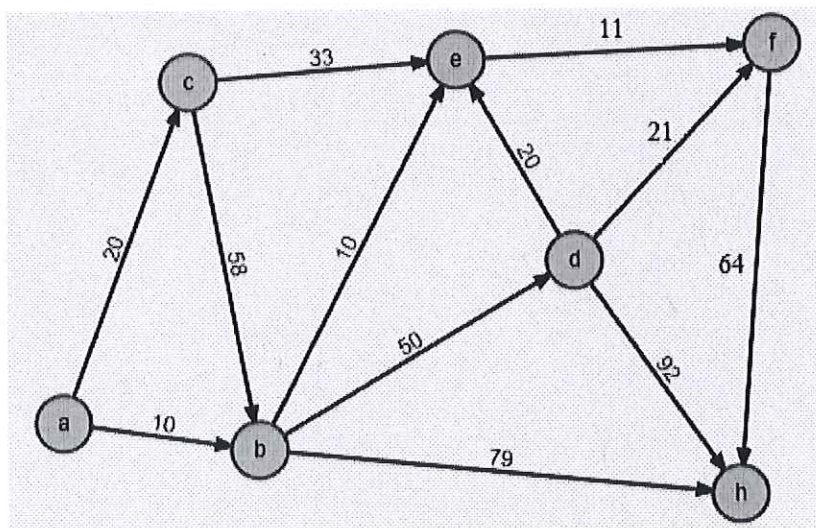
[10]

2. Consider we are constructing Optimal Binary Search Tree. Consider we have given n terms for which it is required to construct OBST.

- a) Write the equation for to compute the cost of the tree. Explain the terms used in this equation.
- b) Write the algorithm to construct OBST and explain what are input and output to this algorithm.

[10]

3. For the following given network, show the stepwise execution of Ford-Fulkerson algorithm. Consider the source node 'a' and sink is 'h'. For each iteration draw the residual graph, write the path for the flow, and computed flow. What is the maximum flow achieved at the end of executing Ford-Fulkerson algorithm?



[10]

4. Consider we have given 7 objects, length of each object is (11,12,13,14,15,16,17)=(3,13,18,9,7,20,11). The maximum capacity of a bin is 22 units. Perform the bin packing using following techniques:

i. Show the bin packing using first fit. How many number of bins required? How many number of bins partially filled?

ii. Show the bin packing using best fit. How many number of bins required? How many number of bins partially filled?

[10]